SEAA 4243 FINITE ELEMENT METHOD · 2L + 1T

Plane Stress
Part 3 of 3

Stresses from displacements, how far to trust them, the 8-node element, how to report FE results — and the project briefing



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Text: Mohd Yassin et al., *Two-Dimensional Finite Element Formulations for Heat, Solids and Fluids (with Matlab)*, MSNM 2020 — §3.4 – §3.5

ACTIVITY 0 · INDIVIDUAL · RETRIEVAL WARM-UP

Three questions from the last session — on paper

5 MIN

Write down

- [k] = t∫[B]ᵀ[E][B] dA — the size of each factor.
- The tip deflection of the 2 × 2 cantilever and of the converged mesh, and the percentage error.
- Why the coarse bilinear mesh is too stiff in bending (one sentence).

WHY Today we compute the stresses that go with those displacements, and ask the same question: how far can they be trusted?

5 minutes. Then the outcomes.



LEARNING OUTCOMES · TODAY

By the end of this session you should be able to

1. Recover strains and stresses at a point of an element from the nodal displacements, $\{\varepsilon\} = [B]\{\hat{u}\}$, $\{\sigma\} = [E][B]\{\hat{u}\}$, and compute principal and von Mises stresses.
2. Explain where stresses are accurate (element centroids, away from supports) and where they are not (element edges, re-entrant corners, point loads).
3. Describe the 8-node element and its advantage per element; convert an edge traction into its 1/6 – 2/3 – 1/6 nodal loads.
4. Plan and report an FE study: idealisation, mesh refinement table, one independent check, quantified conclusions — the project standard.

WHY Outcomes 1–3 complete CLO3. Outcome 4 is CLO4: the project (15%) and the ANSYS demo assess it.



SESSION MAP · 180 MINUTES

Six blocks, six activities, one break, the project briefing

0:00
Stress recovery
35 min
A1 · 8 min

0:35
Accuracy of stresses
25 min
A2 + QR

1:00
The 8-node element
25 min
A3 · 6 min

1:25
Break
10 min
no activity

1:35
Reporting results
25 min
A4 · 8 min

2:00
Project briefing
40 min
A5 · 12 min

2:40
Wrap-up
20 min
Exit ticket

Bring forward

- Plane Stress Part 2 Activity 1: [B] at (1.5, 0.3) with a = 2, b = 1.
- Plane Stress Part 1 Activity 5: the nodal displacements of element 4.
- Plane Stress Part 1 Activity 3: [E] = 224.44, 74.07, 75.19 ($\times 10^6$).

Project

- Groups of [4–5] — lecturer to confirm.
- Brief, deliverables, rubric and dates in Block 5.
- Model plan (one page) drafted in class today.



BLOCK 1 · 0:00 – 0:35

1

Stresses from displacements

The same [B] that built [k] now turns the solved nodal displacements into strains and stresses at any point. Text §3.4, Fig. 3.7.



BLOCK 1 · THE POST-PROCESSING FORMULA

Two matrix products: [B] gives strains, [E] gives stresses

$$\{\varepsilon\}^T = [B]\{\hat{u}\}^T \quad (3\times 8)(8\times 1), \qquad \{\sigma\}^T = [E]\{\varepsilon\}^T = [E][B]\{\hat{u}\}^T$$

Step 1 · locate

Find the element, convert to local coordinates (the local-coordinate rule from heat transfer), take its eight nodal displacements in local order.

Step 2 · strains

Evaluate [B] at the point (3 × 8) and multiply by $\{\hat{u}\}^T$ (8 × 1): three strains.

Step 3 · stresses

Multiply by [E] (3 × 3): σ_{xx}, σ_{yy}, σ_{xy}. Then principal and von Mises if the design needs them.

WHY Software does exactly this at the Gauss points of every element and then averages to the nodes for the contour plot. Hand calculation lets you see what is averaged.



Local (1.5, 0.3): the [B] of Plane Stress Part 2 and the nodal displacements of Plane Stress Part 1

$$[B]_{(1.5,\,0.3)} = \begin{bmatrix} -0.35 & 0 & 0.35 & 0 & 0.15 & 0 & -0.15 & 0 \\ 0 & -0.25 & 0 & -0.75 & 0 & 0.75 & 0 & 0.25 \\ -0.25 & -0.35 & -0.75 & 0.35 & 0.75 & 0.15 & 0.25 & -0.15 \end{bmatrix}$$

$$\{\hat{u}^4\} = \{u_5\ v_5\ \ u_6\ v_6\ \ u_9\ v_9\ \ u_8\ v_8\} \qquad \text{course MATLAB solution of (3.49), six decimals}$$
$$= \{0\ \ -0.043013\ \ 0\ \ -0.130421\ \ 0.041967\ \ -0.130578\ \ 0.030966\ \ -0.045520\}\ \text{m}$$

WATCH OUT [B] is the Plane Stress Part 2 Activity 1 result. {û⁴} is the course MATLAB solution of the text's (3.49) to six decimals: the text prints three, which is not enough here. Local order: nodes (1)(2)(3)(4) of element 4 are global 5, 6, 9, 8, each with u then v.



ACTIVITY 1 · PAIRS · CALCULATION

Strains and stresses at (3.5, 1.3) m

8 MIN

Do

- Multiply row by row: ε_{xx}, ε_{yy}, ε_{xy} = [B]{$\hat{u}^4$}.
- Multiply by [E] (224.44, 74.07, 75.19 × 10^6 N m^{-2}) to get σ_{xx}, σ_{yy}, σ_{xy} in kN m^{-2}.
- Which stress is largest? Is σ_{xx} tension or compression at this point, and should it be (the point is above the mid-height)?

Formula

$\{\varepsilon\}^T = [B]\{\hat{u}\}^T$ $(3 \times 8)(8 \times 1)$

$\{\sigma\}^T = [E]\{\varepsilon\}^T$ $(3 \times 3)(3 \times 1)$

CHECK Only the non-zero products matter: row 1 has two, row 2 four, row 3 six.

8 minutes. Reveal follows.



ACTIVITY 1 · SOLUTION

Tension above the neutral axis; σ_{yy} small, as beam theory assumes

$$
\begin{aligned}
\varepsilon_{xx} &= 0.15(0.041967) - 0.15(0.030966) = 0.0016501 \\
\varepsilon_{yy} &= -0.25(-0.043013) - 0.75(-0.130421) + 0.75(-0.130578) + 0.25(-0.045520) = -0.0007445 \\
\varepsilon_{xy} &= -0.35(-0.043013) + 0.35(-0.130421) + 0.75(0.041967) + 0.15(-0.130578) \\
&\quad +0.25(0.030966) - 0.15(-0.045520) = -0.0041348
\end{aligned}
$$

$$
\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \begin{bmatrix} 224.44 & 74.07 & 0 \\ 74.07 & 224.44 & 0 \\ 0 & 0 & 75.19 \end{bmatrix} \times 10^{6} \begin{Bmatrix} 0.0016501 \\ -0.0007445 \\ -0.0041348 \end{Bmatrix} = \begin{Bmatrix} 315.2 \\ -44.9 \\ -310.9 \end{Bmatrix} \mathrm{kN\,m^{-2}}
$$

WHY $\sigma_{xx} > 0$ above mid-height: top fibre in tension. σ_{xy} carries the end shear; σ_{yy} is small.

WATCH OUT Keep six decimals: v_6 and v_9 differ in the fourth. With three decimals ε_{yy} doubles and σ_{yy} becomes −214.



From three components to the two numbers a design check uses

$$\sigma_{1,2} = \frac{\sigma_{xx} + \sigma_{yy}}{2} \pm \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \sigma_{xy}^2} = 135.2 \pm 359.3$$

$$\Rightarrow \ \sigma_1 = 494, \quad \sigma_2 = -224\ \mathrm{kN\,m^{-2}}$$

$$\sigma_{\mathrm{vM}} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\sigma_{xy}^2} = 637\ \mathrm{kN\,m^{-2}} \qquad (\text{plane stress})$$

Principal stresses

The largest tension (σ_1) governs cracking in concrete; the direction is at ½·atan(2σ_{xy}/(σ_{xx} − σ_{yy})) from the x-axis.

Von Mises

The yield check for steel: σ_{vM} against f_y. ANSYS plots it by default — know what it is before you read the colour.



BLOCK 2 · 0:35 – 1:00

2

How far can a stress be trusted?

Compare with beam theory at the point, at the centroid, across the depth and at the fixed corner. Text Fig. 3.7.



BLOCK 2 · BEAM THEORY AS A CHECK

Beam theory is a valid check only away from the ends

$$\sigma_{xx}^{\text{beam}} = \frac{M\,(y-1)}{I}, \qquad M = P\,(L-x), \qquad P = 10\ \text{kN}, \qquad I = \frac{t\,H^3}{12} = 6.667 \times 10^{-3}\ \text{m}^4$$

Point (m)	2 × 2 FE	Converged FE	Beam theory
(1.0, 1.5) centroid of element 3	1 645	2 265	2 250
(2.0, 1.3) mid-span, on the edge between elements 3 and 4	857 | 185	900	900
(3.0, 1.5) centroid of element 4	519	730	750
(3.5, 1.3) the Activity 1 point, 0.5 m from the loaded end	315	153	225

σ_{xx} in kN m^{-2}; converged = 8-node 64 × 32 from the course solver; at (3.5, 1.3) beam theory gives 5 000 × 0.3 / 6.667 × 10^{-3} = 225.

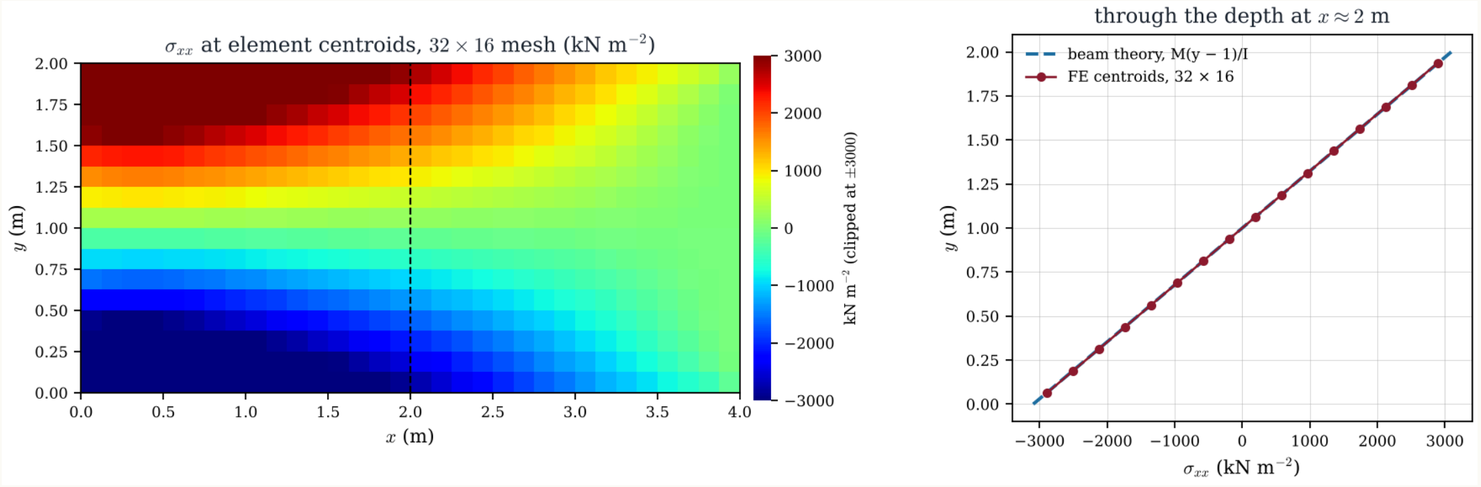
WATCH OUT 2 × 2 → converged is the mesh error: up to 30% at centroids, and two different values on a shared edge.

WHY Converged FE → beam theory: 0.03% at mid-span, 32% at x = 3.5 m. Near the end the uniform traction of the model is not the parabolic shear beam theory assumes (St-Venant).



BLOCK 2 · STRESSES ON A FINE MESH

At element centroids, away from the support, a 32 × 16 mesh matches beam theory



Reading the plot

- Linear through the depth; zero at mid-height: beam theory holds away from the ends.
- Largest at the fixed end, zero at the free end: M = P(L – x).
- Each element has one centroid value; the contour is piecewise constant.

CHECK Centroid values of a 32 × 16 mesh at x ≈ 2 m: within 0.25% of beam theory at every depth.

BLOCK 2 · THE FIXED CORNER NEVER CONVERGES

A re-entrant corner of a fully fixed edge is a stress singularity in the elastic model

Mesh	Centroid of the top-left element (m)	FE σ_{xx} there (kN m^{-2})	Beam theory at the same point	FE / beam
8 × 4	(0.25, 1.75)	4 134	4 219	0.98
16 × 8	(0.125, 1.875)	5 372	5 086	1.06
32 × 16	(0.0625, 1.9375)	6 400	5 537	1.16
64 × 32	(0.031, 1.969)	7 467	5 767	1.30

WATCH OUT Beam theory at the corner tends to 6 000; the FE value passes it and keeps rising with every refinement. The elastic solution is infinite there — refining chases infinity.

WHY Design practice: do not read a peak at a modelled corner. Report stresses a short distance away (St-Venant), or model the support realistically, or use a stress linearisation.



A hole concentrates stress, and the peak converges

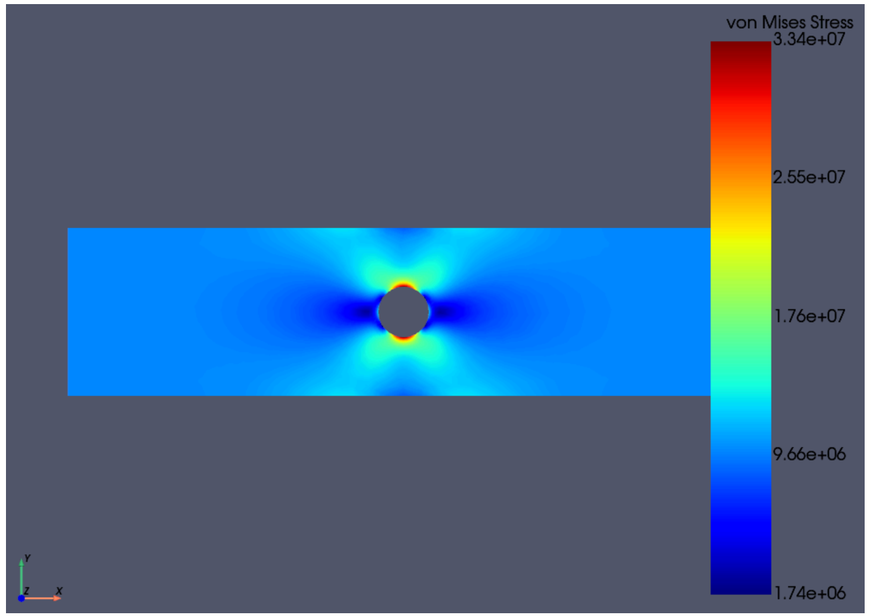


Image: PyMAPDL documentation, “MAPDL 2D plane stress concentration analysis” (PLANE183, plane stress with thickness), © ANSYS, Inc., MIT licence.

The MAPDL example

- Steel plate 0.4 m × 0.1 m in tension, hole d/W = 0.3, 50 elements around the hole.
- Far field 10.0 MPa; peak von Mises 33.4 MPa at the edge of the hole.
- On the net section K_t = 2.34; Peterson’s formula gives 2.35.

CHECK A smooth hole has a finite peak: refine and it converges, so it can be reported once checked — here against Peterson.

WATCH OUT The fixed corner has no finite peak: it grows with every refinement (previous slide). Same plots, opposite verdicts.

ACTIVITY 2 · PAIRS · JUDGEMENT

Which of these stresses would you put in a design report?

6 MIN

Six candidates

- σ_{xx} at the centroid of an element in the middle of a 32 × 16 mesh.
- Peak von Mises at the fixed corner of the same mesh.
- σ_{xx} at the top-right node of the 2 × 2 hand mesh.
- Nodal-averaged σ_{xx} at mid-span from a mesh-converged model.
- Stress under a single nodal point load.
- σ_{xy} averaged over the depth at a section 1 m from the support.

For each

- Trust / use with care / do not report.
- One reason.

6 minutes. Reveal follows.



ACTIVITY 2 · SOLUTION

Trust converged centroid values away from singularities; distribute point loads; never report a corner peak

Candidate	Verdict	Reason
centroid, fine mesh, mid-span	trust	converged, interior point, element-level value
von Mises at the fixed corner	do not report	stress singularity: grows without bound with refinement
nodal σ_{xx} on the 2 × 2 mesh	do not report	coarse mesh, extrapolated to a node
nodal-averaged, converged mesh	trust	averaging is the standard post-processing
under a nodal point load	do not report	another singularity; spread the load over a patch
depth-averaged σ_{xy} 1 m from support	use with care	fine for shear resultant; local values still disturbed by the support



QR EXPLORER · PAIRS, ONE PHONE

Probe the lecture point, then refine

8 MIN

Scan, then do

1. 4-node 2 × 2, probe (3.5, 1.3). Compare with your Activity 1 numbers.
2. Refine to 8-node 16 × 8 and 32 × 16 at the same point. Does σ_{xx} approach beam theory's 225? Then probe mid-span (2.0, 1.3).
3. Run the corner study. Does the peak σ_{xx} at the fixed corner converge?

Stress Recovery Explorer

`razinutm.github.io/My_Lectures/SEAA4243-FEM/labs/stress.html`

8 minutes, optional in class: if the block is running late, set it as homework. The display options are the ones you will meet in ANSYS the next session.



BLOCK 3 · 1:00 – 1:25

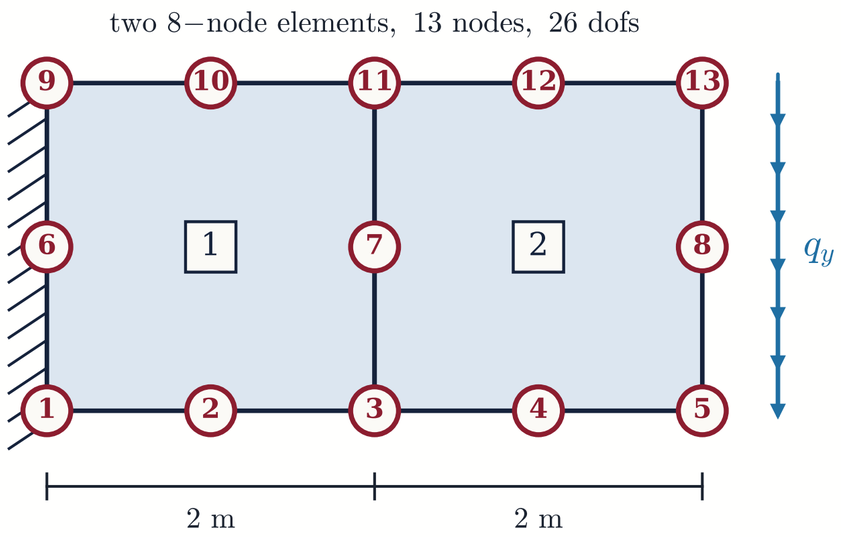
3 The 8-node element

Mid-side nodes, quadratic edges, 16 dofs: fewer elements for the same accuracy in bending. Text Fig. 3.6, (3.50)–(3.53), Table 3.2.



Two elements, 13 nodes, 26 dofs for the whole cantilever

$$[N]:\ 2\times 16, \qquad [B]=[\partial]^T[N]:\ 3\times 16, \qquad [k]=t\iint [B]^T[E][B]\,dA:\ 16\times 16$$



$$N_1 = -\frac{(a-x)(b-y)(2ay-ab+2bx)}{a^2b^2} \qquad \text{corner node}$$

$$N_5 = \frac{4x(a-x)(b-y)}{a^2b} \qquad \text{mid-side node}$$

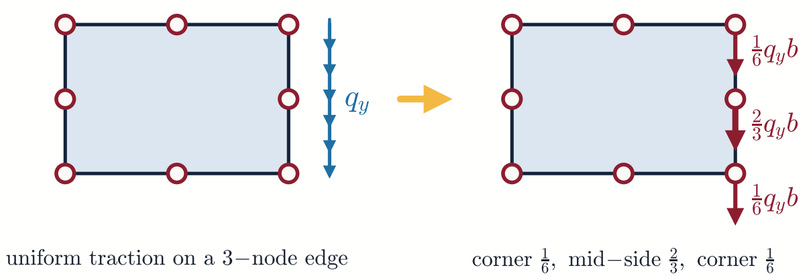
Shape functions of the text's (2.31): corner nodes and mid-side nodes have different forms.

WHY Quadratic along each edge: the element can curve, so bending no longer forces spurious shear. That is the whole advantage.

One sixth to each corner, two thirds to the mid-side node

$$\{r^2\}: \quad \frac{Q_y b}{6} = -1667\ \mathrm{N}, \qquad \frac{2Q_y b}{3} = -6667\ \mathrm{N}, \qquad \frac{Q_y b}{6} = -1667\ \mathrm{N}$$

$$(Q_y = -5000\ \mathrm{N\,m^{-1}},\ b = 2\ \mathrm{m}\text{: one element spans the whole depth})$$



WHY Not half-and-half any more: the quadratic edge shape functions integrate to b/6, 2b/3, b/6. The total is still $q_y b$.

CHECK For the 2 × 1 mesh the loaded edge belongs to element 2 only: U_{10}, U_{16}, U_{26} receive −1667, −6667, −1667 N — the text's (3.52).

ACTIVITY 3 · PAIRS · COMPARISON

Which mesh for the money? Compare Q4 and Q8 on the cantilever

6 MIN

Given · text Tables 3.1 and 3.2

Model	Elements	Dofs	U_{18} (m)
Q4 2 × 2	4	18	−0.131
Q8 2 × 1	2	26	−0.178
Q8 4 × 4	16	—	−0.189
Q4 16 × 16	256	578	−0.189
converged	—	—	−0.192

Find

- The error of each model.
- Elements needed by Q4 and by Q8 for about 1.6% error.
- Why Q8 wins per element — one sentence.

6 minutes. Reveal follows, then the break.



ACTIVITY 3 · SOLUTION · TEXT TABLE 3.2

Two 8-node elements beat four 4-node elements; sixteen match 256

$$\text{Q4 } 2\times 2 \text{ (4 elements, 18 dofs)}: \quad \frac{0.131-0.192}{0.192} = -32\%$$

$$\text{Q8 } 2\times 1 \text{ (2 elements, 26 dofs)}: \quad \frac{0.178-0.192}{0.192} = -7\%$$

$$\text{Q8 } 4\times 4 \text{ (16 elements)}: \quad \frac{0.189-0.192}{0.192} = -1.6\%$$

CHECK ANSYS: PLANE182 ≈ Q4, PLANE183 ≈ Q8. The default for 2D solids is the quadratic one — now you know why.

WHY Cost is dofs, not elements: Q8 4 × 4 has about 130 dofs; Q4 16 × 16 has 578 for the same error.



1:25 – 1:35

Break

Back in 10 minutes. Second half: how to report FE results, and the project briefing.



BLOCK 4 · 1:35 – 2:00

4 Reporting an FE result

A number without a mesh study and a check is an opinion. The four things every results section needs.



BLOCK 4 · FOUR THINGS EVERY FE RESULTS SECTION NEEDS

Idealisation, convergence, an independent check, quantified conclusions

1 · Idealisation stated

Plane stress or plane strain and why; thickness; material constants with units; supports and loads with a sketch; what was left out (self-weight? openings? reinforcement?).

2 · Convergence table

At least three meshes; the quantity of interest for each; the change between the last two as a percentage. One table, one sentence saying which mesh the results come from.

3 · One independent check

Beam theory, equilibrium of reactions against applied loads, a textbook stress-concentration factor, or a hand calculation on a coarse mesh. Report the difference with its sign.

4 · Quantified conclusions

“The tip deflection is 0.191 m, within 0.7% of beam theory” — not “the results agree well”. Say where the maximum is, what it means for the design, and what the model cannot tell you.



BLOCK 4 · THE CANTILEVER WRITTEN UP IN FIVE SENTENCES

A model paragraph for the project report

The cantilever (4 m × 2 m, t = 0.01 m, E = 200 × 10^6 N m^{-2}, ν = 0.33) was idealised as plane stress with the left edge fully fixed and the 5 kN m^{-1} end load applied as an edge traction. Four-node meshes of 2 × 2, 8 × 4, 16 × 8 and 32 × 16 elements gave tip deflections of 0.131, 0.182, 0.189 and 0.191 m; the change between the last two meshes is 1.0%, so the 32 × 16 results are reported. Timoshenko beam theory gives 0.192 m (0.160 m bending, 0.032 m shear), 0.7% above the FE value. The bending stress at element centroids at mid-span agrees with M(y – 1)/I to within 0.25%; the stress at the fixed corner increases without bound under refinement and is not reported. The reactions sum to 10.0 kN upward and 40.0 kN m, equal to the applied load and its moment.

CHECK Every sentence carries a number, a comparison or a decision. No adjectives.

WHY Five sentences, one table, two figures: that is the whole results section of a good project report.



ACTIVITY 4 · PAIRS · WRITING

Write the heat-transfer plate up in three sentences

8 MIN

Data · Heat Transfer Part 2

Mesh	T(L, 0) (°C)
Q4 2 × 2	0.6214
Q4 16 × 16	0.5898
Software	0.5897

Plate 1 m × 1 m, k = 300 W m^{-1} K^{-1}, q_h = 600 W m^{-3}, two edges at 0 °C, two insulated.

Write

- Sentence 1: the idealisation.
- Sentence 2: the convergence, with the percentage change.
- Sentence 3: the check against software, with the signed difference.

Rule: every sentence has at least one number.

8 minutes. Two pairs read theirs out.



BLOCK 5 · 2:00 – 2:40

5

Project briefing

CLO4: investigate a structural problem with FEM, in a group, with software — and defend the numbers. 15% of the course.



BLOCK 5 · THE BRIEF

One real structure, one FE model, one verified answer, one report, one presentation

Task

Choose a 2D structural problem, idealise it, model it in ANSYS (plane stress or plane strain), verify the model, and interpret the results for design.

Team

Groups of [4–5]. One submission per group; a one-line statement of who did what.

Weight

15% of the course (CLO4). Report [70%] · presentation [30%] — lecturer to confirm.

Deliverables

- Technical report, [10–15] pages: idealisation, mesh study, check, results, interpretation, limitations.
- Presentation, [10] minutes + questions, [date].
- Model files (.wbpj or archive) and the hand-check sheet.

Dates

- Today: groups formed, model plan drafted.
- ANSYS demo: bring your plan and start the model in the lab.
- [date]: report and presentation.



BLOCK 5 · CANDIDATE STRUCTURES

Four families that fit the course; propose your own if it is 2D and checkable

Shear wall with openings

Text Exercise 3.4: 5 m × 6 m, t = 0.2 m, 125 kN m^{-1} on top, two openings. Plane stress. Check: cantilever beam + stress concentration at opening corners.

Deep beam or transfer girder

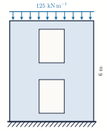
Span-to-depth about 2; point or distributed load; supports on bearing pads. Plane stress. Check: beam theory for deflection, strut-and-tie for the stress path.

Gusset or connection plate

Steel plate with bolt-hole or weld loads; t = 10–20 mm. Plane stress. Check: net-section stress, stress-concentration factor at the hole (≈ 3).

Dam or retaining wall section

Per metre run with hydrostatic pressure and self-weight. Plane strain. Check: overturning and base stress from statics.



WATCH OUT Choose something with a known hand check. A structure you cannot check is a structure you cannot defend in the presentation.

BLOCK 5 · HOW IT IS MARKED

Proposed rubric — lecturer to confirm weights

Criterion	Weight	Full marks look like	Common failure
Idealisation and modelling	30%	plane stress/strain justified; supports and loads realistic; units consistent; sketch of the model	fixing edges that are free; point loads at a node; wrong thickness
Verification	30%	three-mesh convergence table; one independent hand check with a signed difference	single mesh; “results agree well” with no number
Interpretation	20%	where the maximum is and why; what it means for the design; singularities identified and excluded	quoting the corner peak as the design stress
Report and presentation	20%	five-sentence results paragraph; labelled figures with units; questions answered with numbers	screenshots without scales; reading slides aloud

WHY Sixty per cent of the marks are for modelling decisions and verification — not for the colour plots. A plain model that is checked beats an elaborate one that is not.



BLOCK 5 · THE MODELLING WORKFLOW

Eight steps; the project report follows the same order

1 Idealise
2D? stress or strain? what is left out

2 Geometry
dimensions, thickness, openings

3 Material
E, ν, units

4 Mesh
element type, size, three levels

5 Supports
fixed, roller, symmetry

6 Loads
tractions, body force, not point loads

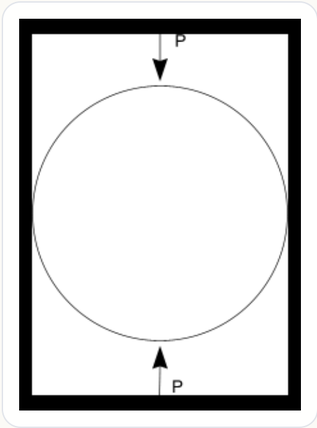
7 Solve & verify
reactions, convergence, hand check

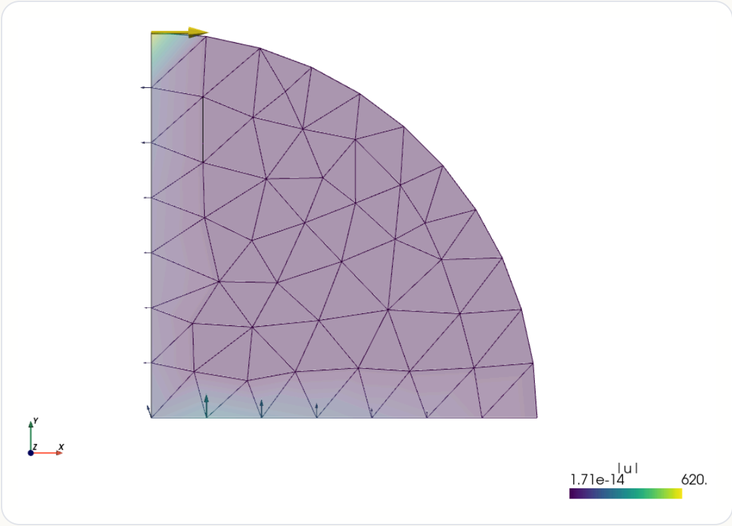
8 Interpret
where, why, what it means

CHECK the ANSYS demo runs exactly this sequence in ANSYS on the cantilever and the plate, then on your structure.



Use symmetry, and do not read stresses under a point load





Indirect tensile (Brazilian) test

- A 150 mm disk squeezed across its diameter by two loads P; plane strain.
- Two lines of symmetry → model one quarter: rollers on the cut edges (no displacement normal to them).
- The load is a point force: the stress under it is singular. Read results away from it, or spread it over the real contact width.

Images: SfePy documentation, Primer (indirect tensile test in 2D), BSD-3-Clause licence, sfepy.org.

ACTIVITY 5 · GROUPS · PLANNING

Draft your model plan on one page

12 MIN

Write, as a group

- The structure, with a sketch and dimensions; the source of the geometry and loads.
- Idealisation: plane stress or plane strain, thickness, material (E, ν, units).
- Supports and loads, each one justified in a few words.
- The quantity you will report (deflection? stress where?) and the hand check you will use.
- Three mesh sizes you will run.

WATCH OUT Lecturers circulate. A plan with a fixed edge that should be free, or no hand check, is sent back now rather than in the report.

CHECK Hand in the plan at the end; it comes back with comments at the ANSYS session.

12 minutes. Hand in before you leave.



BLOCK 6 · 2:40 – 3:00

6 Wrap-up

Three sessions of plane stress in one page, the exit ticket, and what to bring to the ANSYS session.



BLOCK 6 · PLANE STRESS IN ONE PAGE

From equilibrium to a verified stress

Part	Step	Result
1	equilibrium + Hooke + strain–displacement	$[\partial][E][\partial]^T\{u\}^T = -\{f\}$; traction natural, displacement essential; [N] 2 × 8
2	Galerkin + IBP; $[B] = [\partial]^T[N]$	$[k] = t\int[B]^T[E][B]\,dA$, $s_1 \ldots s_{12}$; loads qb/2; assembly by dof map; 2 × 2 cantilever 32% too stiff
3	post-processing and judgement	$\{\sigma\} = [E][B]\{\hat{u}\}$; trust centroids on converged meshes; corners are singular; Q8 curves

WHY The heat-transfer pipeline, with two unknowns per node. Nothing else changed.

CHECK Exam: expect one hand calculation on a single element (k entry, load vector or stress at a point) and one interpretation question.



ACTIVITY 6 · INDIVIDUAL · EXIT TICKET

Three questions before you leave

5 MIN

Questions

- Order the three operations that take nodal displacements to a stress at a point.
- Your model shows its highest stress at a fixed corner. What do you do?
- Which element would you choose for a wall in bending with a limited number of elements, Q4 or Q8, and why?

Five minutes. Name and matric number; for feedback, not marks.



WRAP-UP · BEFORE THE ANSYS SESSION

Bring the model plan, the two textbook examples and a laptop with ANSYS access

Today in four lines

- {σ} = [E][B]{û}: two products; use centroid values on converged meshes.
- Stresses converge slower than displacements; fixed corners and point loads are singular.
- Q8: quadratic edges, 1/6–2/3–1/6 loads, far fewer elements for the same error.
- A results section = idealisation + convergence table + one check + numbers.

Before the ANSYS demo

- Install or confirm access to ANSYS Student / Workbench; bring a laptop.
- Re-read Heat Transfer Part 2 §2.3 and Plane Stress Part 2 §3.4: we rebuild both in ANSYS and compare.
- Finish the group model plan if it was returned with comments.

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